

Boundary Value Problems and Sturm–Liouville Theory

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb from telling an IVP from a BVP and solving simple two-point problems, through the full three-case eigenvalue analysis and the shooting method, up to the orthogonality integral, a complete eigenfunction expansion with every coefficient integral expanded, and the Sturm–Liouville orthogonality proof. Solutions follow in Part III.

1. **(IVP vs. BVP)** State what distinguishes a boundary value problem from an initial value problem, and identify which of $y(0) = 1, y'(0) = 2$ versus $y(0) = 1, y(2) = 5$ is which.
2. **(Simple Dirichlet BVP)** Solve $y'' = 0$ on $[0, 2]$ with $y(0) = 1, y(2) = 5$.
3. **(Nonhomogeneous BVP)** Solve $y'' = 2$ on $[0, 1]$ with $y(0) = 0, y(1) = 0$.
4. **(Infinitely many solutions)** Show that $y'' + y = 0$ with $y(0) = 0, y(\pi) = 0$ has infinitely many solutions, and give their form.
5. **(Eigenvalues, three cases)** For $y'' + \lambda y = 0, y(0) = y(L) = 0$, carry out the $\lambda < 0, \lambda = 0, \lambda > 0$ analysis and state λ_n and y_n .
6. **(Neumann spectrum)** Find the eigenvalues and eigenfunctions of $y'' + \lambda y = 0$ with $y'(0) = y'(L) = 0$.
7. **(Shooting method)** Solve $y'' = y$ on $[0, 1], y(0) = 0, y(1) = 1$ by two shots and linear interpolation.
8. **(Orthogonality integral)** Evaluate $\int_0^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx$ for $m \neq n$ and for $m = n$.
9. **(Eigenfunction expansion)** Expand $f(x) = x$ on $[0, L]$ in the sine series $\sum_{n \geq 1} c_n \sin \frac{n\pi x}{L}$, computing c_n in full.
10. **(Sturm–Liouville capstone)** Write $X'' + \lambda X = 0, X(0) = X(L) = 0$ in S–L form (name p, q, w), then prove $\int_a^b y_n y_m w dx = 0$ for $\lambda_n \neq \lambda_m$.

Part III — Complete Solutions

Solution 1. A BVP imposes its side conditions at *two different points*, whereas an IVP stacks all of them at a single point. Hence

$$y(0) = 1, \quad y'(0) = 2 \text{ (both at } x = 0) \Rightarrow \text{IVP}; \quad y(0) = 1, \quad y(2) = 5 \text{ (at } x = 0, 2) \Rightarrow \text{BVP}.$$

Unlike the IVP, the BVP carries no automatic existence–uniqueness guarantee.

Solution 2. The general solution of $y'' = 0$ is $y = c_1 + c_2x$. Imposing the endpoints,

$$y(0) = c_1 = 1; \quad y(2) = 1 + 2c_2 = 5 \Rightarrow c_2 = 2.$$

Unique solution: $y(x) = 1 + 2x$.

Solution 3. Integrate twice: $y = x^2 + c_2x + c_1$. Then

$$y(0) = c_1 = 0; \quad y(1) = 1 + c_2 = 0 \Rightarrow c_2 = -1.$$

Unique solution: $y(x) = x^2 - x$.

Solution 4. The general solution is $y = A \cos x + B \sin x$. The conditions give

$$y(0) = A = 0; \quad y(\pi) = A \cos \pi + B \sin \pi = -A = 0,$$

which is satisfied for *every* B . Thus $y = B \sin x$ is a solution for all B — infinitely many (here $\lambda = 1$ is an eigenvalue).

Solution 5. For $\lambda < 0$ and $\lambda = 0$ the BCs force $y \equiv 0$. For $\lambda = \mu^2 > 0$, $y = A \cos \mu x + B \sin \mu x$ with $y(0) = A = 0$, so $y = B \sin \mu x$ and $y(L) = B \sin \mu L = 0$:

$$\sin \mu L = 0 \Rightarrow \mu L = n\pi \Rightarrow \lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad y_n = \sin \frac{n\pi x}{L}, \quad n = 1, 2, 3, \dots$$

Solution 6. With $y = A \cos \mu x + B \sin \mu x$, $y' = \mu(-A \sin \mu x + B \cos \mu x)$. Then $y'(0) = \mu B = 0 \Rightarrow B = 0$, and $y'(L) = -\mu A \sin \mu L = 0$:

$$\sin \mu L = 0 \Rightarrow \lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad y_n = \cos \frac{n\pi x}{L}, \quad n = 0, 1, 2, \dots$$

The Neumann case keeps $n = 0$: $\lambda_0 = 0$ with constant eigenfunction $y_0 = 1$.

Solution 7. Recast as $y'' = y$, $y(0) = 0$, $y'(0) = s$, with exact IVP solution $y(x; s) = s \sinh x$. Two shots:

$$F(0) = 0 - 1 = -1; \quad F(1) = \sinh 1 - 1 \approx 0.1752; \quad s^* = \frac{1}{\sinh 1} \approx 0.8509.$$

Final solution: $y(x) = \frac{\sinh x}{\sinh 1}$, exact since the problem is linear.

Solution 8. Use $\sin \alpha \sin \beta = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)]$. For $m \neq n$ both cosine integrals vanish over $[0, L]$, giving 0. For $m = n$,

$$\int_0^L \sin^2 \frac{n\pi x}{L} dx = \int_0^L \frac{1 - \cos \frac{2n\pi x}{L}}{2} dx = \frac{L}{2}.$$

So the integral is 0 for $m \neq n$ and $L/2$ for $m = n$.

Solution 9. With weight $w = 1$, $c_n = (\int_0^L x \sin \frac{n\pi x}{L} dx) / (\int_0^L \sin^2 \frac{n\pi x}{L} dx)$. The denominator is $L/2$; integrating the numerator by parts,

$$\int_0^L x \sin \frac{n\pi x}{L} dx = \left[-\frac{Lx}{n\pi} \cos \frac{n\pi x}{L} \right]_0^L + \frac{L}{n\pi} \int_0^L \cos \frac{n\pi x}{L} dx = (-1)^{n+1} \frac{L^2}{n\pi} + 0.$$

$$c_n = \frac{(-1)^{n+1} L^2 / (n\pi)}{L/2} = (-1)^{n+1} \frac{2L}{n\pi} \Rightarrow x = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2L}{n\pi} \sin \frac{n\pi x}{L}.$$

Solution 10. Matching $(pX')' + (q + \lambda w)X = 0$ to $X'' + \lambda X = 0$ gives $p = 1$, $q = 0$, $w = 1$ (regular, since $p, w > 0$ with separated BCs). For the proof, multiply $(py'_n)' + \lambda_n w y_n = 0$ by y_m , the m -equation by y_n , subtract, and integrate:

$$(\lambda_n - \lambda_m) \int_a^b w y_n y_m dx = [p(y_n y'_m - y_m y'_n)]_a^b = 0,$$

where the bracket vanishes by the homogeneous BCs. Since $\lambda_n \neq \lambda_m$, $\int_a^b y_n y_m w dx = 0$.

Unit Key Takeaway

One thread ties the unit together: **boundary conditions create a spectrum, and orthogonality turns that spectrum into a basis**. Splitting the side data across two endpoints can leave a linear problem with no solution, one, or infinitely many — the last case is where the **eigenvalues** $\lambda_n = (n\pi/L)^2$ and **eigenfunctions** $\sin(n\pi x/L)$ live. Those eigenfunctions are **orthogonal with respect to a weight** w , so any reasonable f expands as $f = \sum c_n y_n$ with each $c_n = \langle f, y_n \rangle / \langle y_n, y_n \rangle$ a single integral. The **Sturm–Liouville form** $(py')' + (q + \lambda w)y = 0$ is the master frame that guarantees real eigenvalues, weighted orthogonality, and a complete functional basis.